

Pandrosion Relativity

Finite-slope causal probes, dynamic atlases, and geodesic physical motion

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Abstract

This paper develops a “Pandrosion relativity” formulation. It does not replace special or general relativity. Instead, it rewrites the relativistic motion law as a finite-slope Pandrosion probe calculus: admissible probes must stay inside the future light cone, massive-particle paths are selected by a discrete proper-time score, and dynamic atlases remove coordinate defects in the same spirit as Thales scaling and Riemann-chart inversion in monomial Pandrosion.

Let $(\mathcal{M}, g)$ be a time-oriented Lorentzian manifold. A Pandrosion worldline is a causal graph path

$$\Gamma = (p_0, p_1, \dots, p_N), \quad p_{k+1} \in \mathcal{C}_h^+(p_k),$$

with score

$$\begin{aligned} \mathcal{J}_h(\Gamma) = & \sum_{k=0}^{N-1} \tau_h(p_k, p_{k+1}) - \lambda_h \sum_k \mathcal{A}_h(p_{k-1}, p_k, p_{k+1}) \\ & - \mu_h \sum_k \mathcal{K}_h(p_k, p_{k+1}) - \nu_h \sum_k \Theta_h(p_k, p_{k+1}). \end{aligned}$$

Here τ_h is the finite-segment proper time, $\mathcal{A}_h$ penalizes finite-slope acceleration, $\mathcal{K}_h$ penalizes curvature or tidal stiffness, and Θ_h penalizes ill-conditioned charts. The main theorem states that, on a compact globally hyperbolic domain and under causal graph consistency, maximizers of $\mathcal{J}_h$ converge to proper-time maximizing causal geodesics as $h \rightarrow 0$, when the penalty weights vanish at a consistent rate. A second result shows that the dynamic-atlas formulation is chart invariant in the continuum limit. Finally, the weak-field slow-motion expansion recovers Newton’s law, while a finite-holonomy conjecture gives the natural Pandrosion route toward Einstein’s field equation.

One-sentence summary. Pandrosion relativity is a finite-probe version of relativistic geodesic motion: replace derivatives by causal finite slopes, replace one coordinate system by a dynamic atlas, and recover standard relativity in the continuum limit.

Status. The graph-convergence and chart-covariance statements are mathematical approximation theorems under explicit consistency hypotheses. The finite curvature-to-field-equation step is stated as a research conjecture, not as a new proof of general relativity.

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1 Purpose of the paper

The previous Pandrosion papers built three ideas that naturally touch relativity:

- (i) Paper 000 showed that a useful computation may require a chart choice, a homothety, and sometimes a reciprocal Riemann chart.
- (ii) Papers 011–013 treated probes as geodesic objects selected by a stability metric.
- (iii) Paper 019 proved that dense finite probe graphs can converge to continuous geodesic actions.

Relativity is also a geometric theory. The free motion of a massive particle is not defined by a preferred coordinate acceleration. It is defined by a causal path that extremizes proper time and is independent of the chart used to describe it.

This makes a clean Pandrosion translation possible. The derivative of a worldline is replaced by finite causal segments. The Christoffel-symbol geodesic equation is replaced by a finite-slope transport condition. Bad coordinates are treated as chart defects, not as physical singularities. The resulting method is not a new physical law; it is a finite-probe language for the existing relativistic law.

2 Relativistic Pandrosion state

Let $(\mathcal{M}, g)$ be a smooth time-oriented Lorentzian manifold of dimension $d + 1$, with signature $(- + \cdots +)$. We use units $c = 1$ until the Newtonian limit section. A future-directed tangent vector v is timelike when

$$g_p(v, v) < 0.$$

The proper time of a timelike C^1 curve γ is

$$\mathcal{T}(\gamma) = \int \sqrt{-g_{\gamma(t)}(\dot{\gamma}(t), \dot{\gamma}(t))} \, dt.$$

Definition 2.1 (Pandrosion relativistic state). *A relativistic Pandrosion state is*

$$\mathfrak{s} = (p, i, e, \rho),$$

where $p \in \mathcal{M}$ is the current event, i is the active chart or tetrad index, e is a local future timelike frame vector, and $\rho > 0$ is the current probe scale.

The scale ρ is the relativistic analog of Thales normalization. It keeps probe segments inside a local region where the metric, chart transition, and finite-slope estimates are stable. The chart index i is the analog of the direct or reciprocal Riemann chart in Paper 000: it is not physics, but it controls numerical stability.

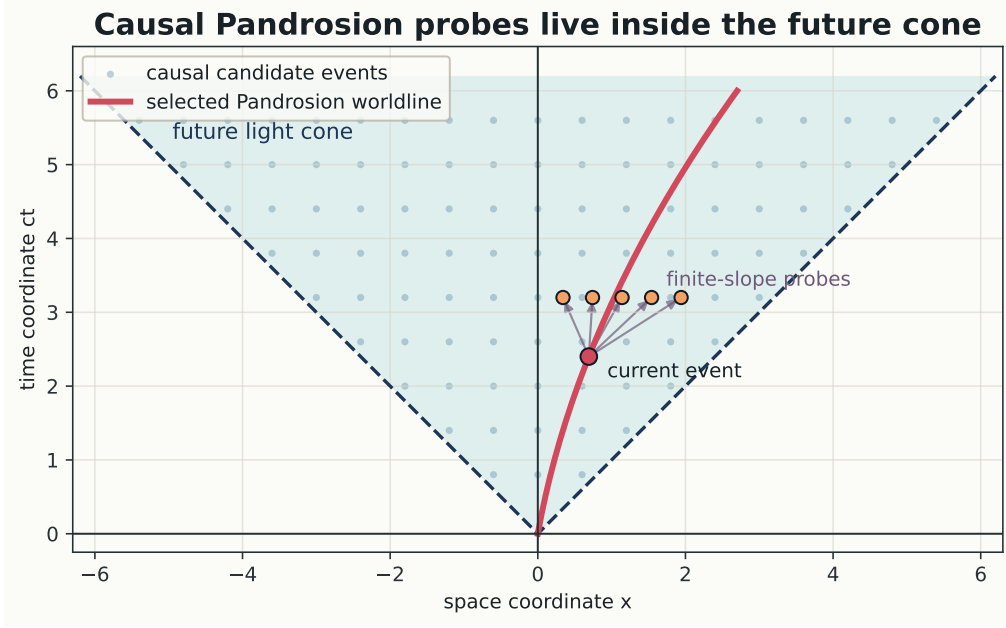


Figure 1: Causal Pandrosion probe geometry. The candidate endpoint set is not Euclidean; it is restricted by the future light cone. A worldline is selected from finite causal probes by a proper-time score plus finite-slope and atlas penalties.

Definition 2.2 (Causal probe set). *Let $X_h \subset \mathcal{M}$ be a finite event set with mesh size h . The future causal Pandrosion probe set at $p \in X_h$ is*

$$\mathcal{C}_h^+(p) = \{q \in X_h : q \in J^+(p), h_- \leq \tau_h(p, q) \leq h_+\},$$

where $J^+(p)$ is the causal future of p , $0 < h_- \leq h_+$, and $\tau_h(p, q)$ is a local finite-segment proper-time estimator.

In a chart ϕ_i , write $\Delta x^\alpha = x^\alpha(q) - x^\alpha(p)$. For sufficiently small segments one may use

$$\tau_{h,i}(p, q) = \sqrt{\max\{0, -g_{\alpha\beta}^{(i)}(p) \Delta x^\alpha \Delta x^\beta\}}.$$

Higher-order quadrature may replace this first-order estimator. The important point is that the endpoint is admissible only if the finite segment is future-causal.

3 Finite-slope proper-time score

A discrete Pandrosion worldline is a graph path

$$\Gamma = (p_0, p_1, \dots, p_N), \quad p_{k+1} \in \mathcal{C}_h^+(p_k).$$

The raw discrete proper time is

$$\mathcal{T}_h(\Gamma) = \sum_{k=0}^{N-1} \tau_h(p_k, p_{k+1}).$$

For a free massive particle, the continuum law says that the physical path is a proper-time extremal. In a convex normal neighborhood this is a local maximum of proper time among nearby timelike curves with fixed endpoints. Pandrosion therefore uses a maximizing score, not a descent score:

$$\mathcal{J}_h(\Gamma) = \mathcal{T}_h(\Gamma) - \mathcal{P}_h(\Gamma).$$

The penalty $\mathcal{P}_h$ contains the terms that make finite probes stable.

Definition 3.1 (Pandrosion relativistic score). For $\Gamma = (p_0, \dots, p_N)$, set

$$\begin{aligned} \mathcal{J}_h(\Gamma) = & \sum_{k=0}^{N-1} \tau_h(p_k, p_{k+1}) - \lambda_h \sum_{k=1}^{N-1} \mathcal{A}_h(p_{k-1}, p_k, p_{k+1}) \\ & - \mu_h \sum_{k=0}^{N-1} \mathcal{K}_h(p_k, p_{k+1}) - \nu_h \sum_{k=0}^{N-1} \Theta_h(p_k, p_{k+1}). \end{aligned}$$

Here $\mathcal{A}_h$ is a finite-slope acceleration penalty, $\mathcal{K}_h$ is a curvature or tidal-stiffness penalty, and Θ_h is a chart-conditioning penalty.

The penalty terms are not external forces. They are numerical stabilizers. For a continuum convergence theorem, their weights should vanish as $h \rightarrow 0$. For a practical engine, they prevent a finite graph from choosing segments that are formally causal but unstable because of a poor chart, a large tidal gradient, or a noisy finite-slope estimate.

3.1 Finite-slope connection

Let $g_{\alpha\beta}$ be sampled on a local event lattice. A finite difference $D_\mu^h g_{\alpha\beta}(p)$ gives the finite-slope Christoffel symbol

$$\Gamma_{h,\alpha\beta}^\mu(p) = \frac{1}{2} g_h^{\mu\nu}(p) \left(D_\alpha^h g_{\beta\nu}(p) + D_\beta^h g_{\alpha\nu}(p) - D_\nu^h g_{\alpha\beta}(p) \right).$$

For a graph path, define segment velocities

$$V_k^\mu = \frac{x^\mu(p_{k+1}) - x^\mu(p_k)}{\Delta t_k}.$$

A finite-slope geodesic defect is then

$$E_{h,k}^\mu = \frac{V_{k+1}^\mu - V_k^\mu}{\Delta t_k} + \Gamma_{h,\alpha\beta}^\mu(p_k) V_k^\alpha V_k^\beta.$$

The acceleration penalty may be chosen as

$$\mathcal{A}_h(p_{k-1}, p_k, p_{k+1}) = \sum_\mu |E_{h,k}^\mu|^2 \Delta t_k.$$

Thus the usual differential equation

$$\ddot{x}^\mu + \Gamma_{\alpha\beta}^\mu \dot{x}^\alpha \dot{x}^\beta = 0$$

is recovered as the zero-defect limit of finite-slope probes.

4 Main convergence theorem

The next theorem is the relativistic analog of the graph-action theorem in Paper 019. The sign changes because timelike geodesics maximize proper time locally, while the Riemannian stability actions in Paper 019 minimized length.

Assumption 4.1 (Causal graph consistency). Let $\mathcal{D} \subset \mathcal{M}$ be a compact causally convex subset of a globally hyperbolic spacetime, and let $p_-, p_+ \in \mathcal{D}$ be timelike related. A family of finite event graphs $X_h \subset \mathcal{D}$ is causally consistent if:

- (i) the mesh of X_h in $\mathcal{D}$ tends to zero;
- (ii) graph edges are future-causal and have maximum coordinate size $\mathcal{O}(h)$;

(iii) every smooth future-timelike curve in $\mathcal{D}$ can be uniformly sampled by future-causal graph paths;

(iv) the segment estimator satisfies

$$\tau_h(a, b) = \tau_{\text{geo}}(a, b) + \mathcal{O}(h^3)$$

uniformly for small timelike segments, where $\tau_{\text{geo}}(a, b)$ is the geodesic proper time between nearby events.

Assumption 4.2 (Vanishing stabilizers). *The penalties in $\mathcal{J}_h$ are nonnegative and satisfy*

$$\sup_{\Gamma \in \mathcal{G}_h} \left[\lambda_h \sum \mathcal{A}_h + \mu_h \sum \mathcal{K}_h + \nu_h \sum \Theta_h \right] \rightarrow 0$$

on every uniformly bounded family of consistent timelike graph paths $\mathcal{G}_h$.

Theorem 4.3 (Pandrosion causal graph convergence). *Assume causal graph consistency and vanishing stabilizers. Let Γ_h maximize $\mathcal{J}_h$ among future-causal graph paths from nodes converging to p_- to nodes converging to p_+ . If the polygonal interpolants of Γ_h have a uniformly convergent subsequence, then every subsequential limit γ is a proper-time maximizing future-causal curve from p_- to p_+ . On a smooth timelike interior where the maximizer is regular, γ is a timelike geodesic. If the maximizing geodesic is unique, the full family Γ_h converges to it.*

Proof. Let

$$\mathcal{T}_*(p_-, p_+) = \sup_{\eta} \mathcal{T}(\eta),$$

where the supremum is over future-causal curves from p_- to p_+ in $\mathcal{D}$. By global hyperbolicity and compactness of $\mathcal{D}$, maximizing causal curves exist.

For the lower bound, take a smooth timelike curve η whose proper time is within ε of $\mathcal{T}_*$. Causal graph consistency supplies graph paths Γ_h^η converging uniformly to η . The segment estimator is consistent, so

$$\lim_{h \rightarrow 0} \mathcal{T}_h(\Gamma_h^\eta) = \mathcal{T}(\eta).$$

The stabilizers vanish on these paths, hence

$$\liminf_{h \rightarrow 0} \sup_{\Gamma} \mathcal{J}_h(\Gamma) \geq \mathcal{T}(\eta) \geq \mathcal{T}_* - \varepsilon.$$

Letting $\varepsilon \rightarrow 0$ gives the recovery inequality.

For the upper bound, let Γ_{h_j} be a maximizing subsequence whose interpolants converge uniformly to γ . Lorentzian length is upper semicontinuous for uniformly convergent causal curves in a globally hyperbolic domain, and the local segment estimator differs from geodesic proper time by an error summing to $o(1)$. Since the penalties are nonnegative,

$$\limsup_{j \rightarrow \infty} \mathcal{J}_{h_j}(\Gamma_{h_j}) \leq \limsup_{j \rightarrow \infty} \mathcal{T}_{h_j}(\Gamma_{h_j}) \leq \mathcal{T}(\gamma) \leq \mathcal{T}_*.$$

Together with the lower bound, $\mathcal{T}(\gamma) = \mathcal{T}_*$, so γ maximizes proper time. A regular timelike proper-time extremal satisfies the geodesic equation. Uniqueness removes the need to pass to subsequences. $\square$

Corollary 4.4 (Finite-slope geodesic equation). *Under the hypotheses of Theorem 4.3, if the finite metric differences $D^h g$ converge uniformly to ∂g on the limiting tube, then the finite-slope defects $E_{h,k}^\mu$ converge weakly to zero along any regular interior segment of the limiting worldline.*

Proof. The graph maximizers converge to a regular timelike geodesic. Uniform convergence of $D^h g$ to ∂g gives convergence of Γ_h to the Levi-Civita symbols. The discrete Euler condition for nearby probe variations is therefore the finite-difference form of the geodesic Euler–Lagrange equation, with consistency error tending to zero. $\square$

5 Dynamic atlas and horizon-safe charting

Coordinate singularities are a central reason to use an atlas rather than a single chart. The Schwarzschild radial coordinate is a standard example: the coordinate expression has a singularity at $r = 2M$, but the horizon is not a curvature singularity. A Pandrosion engine should treat this as a chart conditioning problem.

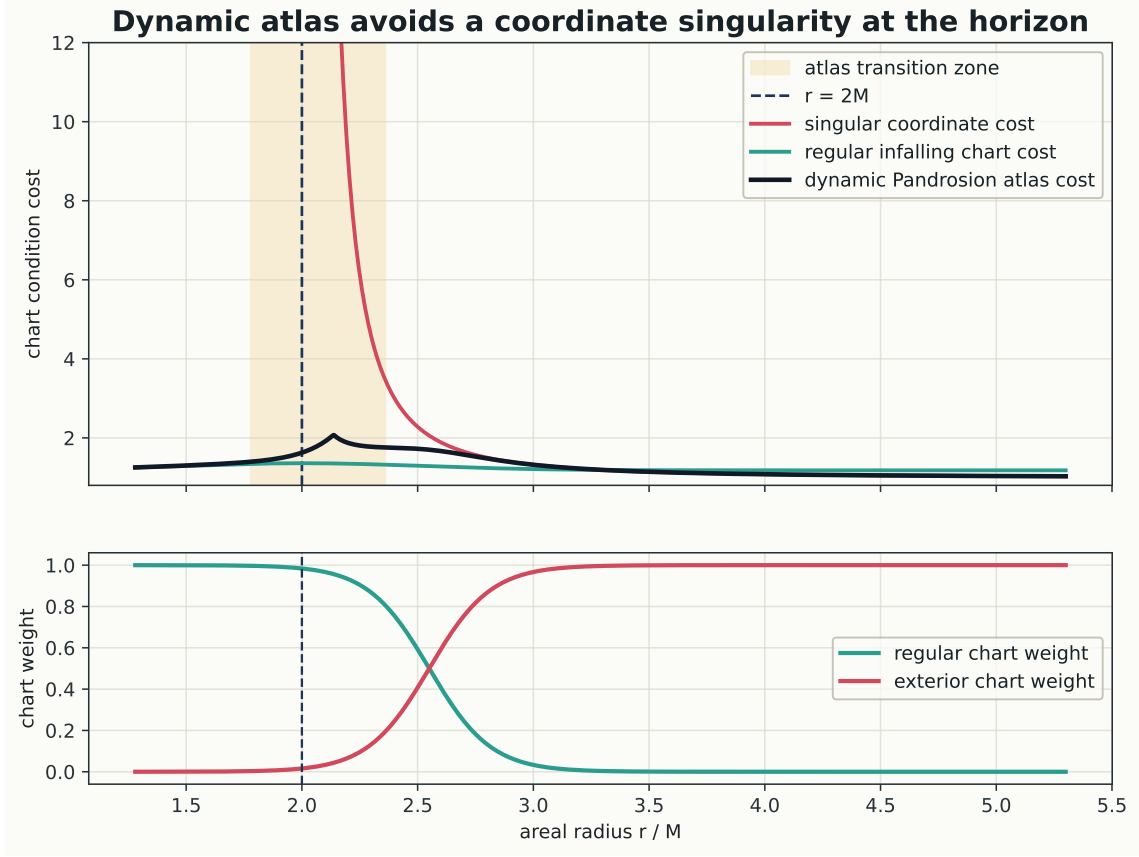


Figure 2: Dynamic chart selection near a horizon. A coordinate chart may become ill-conditioned even when the spacetime geometry is regular. The Pandrosion atlas switches to a regular infalling chart near the bad coordinate region and returns to the exterior chart away from it.

Let $\{(U_i, \phi_i)\}_{i \in I}$ be a smooth atlas on $\mathcal{D}$. A charted worldline stores

$$(p_0, i_0), (p_1, i_1), \dots, (p_N, i_N).$$

For an edge (p_k, p_{k+1}) , the local segment time is evaluated in chart i_k , while the transition $i_k \rightarrow i_{k+1}$ is charged a small atlas cost

$$\Theta_h(p_k, p_{k+1}, i_k, i_{k+1}) = \theta_i(p_k) + \theta_j(p_{k+1}) + \sigma_h \left\| D\phi_j \circ D\phi_i^{-1} \right\|.$$

Here θ_i measures chart conditioning. It is allowed to grow near a coordinate defect, forcing the optimizer to select a better chart.

Proposition 5.1 (Local chart covariance). *Let two charts i, j overlap on a compact region and have bounded derivatives through order three. If a segment has size h , then the first-order finite-segment proper-time estimators computed in the two charts satisfy*

$$\tau_{h,i}(p, q) - \tau_{h,j}(p, q) = \mathcal{O}(h^3).$$

Consequently, along a path with $\mathcal{O}(h^{-1})$ segments and bounded total time, the total chart discrepancy is $\mathcal{O}(h^2)$.

Proof. The exact quadratic expression $-g_p(v, v)$ is tensorial. The only difference between chart estimators comes from representing the finite displacement $q - p$ by first-order chart increments instead of the exact exponential displacement. Taylor expansion of the chart transition map gives an error of order h^3 in the segment proper time on compact overlaps. Summing over $\mathcal{O}(h^{-1})$ segments gives $\mathcal{O}(h^2)$. $\square$

Theorem 5.2 (Dynamic-atlas invariance). *Assume the hypotheses of Theorem 4.3 and let the atlas transition weights vanish while still excluding charts whose condition cost diverges. Then the limiting Pandrosion worldline is independent of the chosen smooth atlas. Different atlases may choose different finite graph paths, but every convergent maximizing family has the same continuum proper-time maximizer when that maximizer is unique.*

Proof. On every compact overlap the segment-time discrepancy between smooth charts is summable to $o(1)$ by the chart-covariance proposition. Divergent chart costs only remove ill-conditioned coordinate descriptions; they do not remove the underlying event from the manifold if another regular chart covers it. Therefore all admissible smooth atlases produce the same limiting proper-time functional. The causal graph convergence theorem then applies. $\square$

6 Relation to Thales scaling and Riemann inversion

The terminology in Paper 000 can be translated almost literally.

Monomial Pandrosion	Relativistic Pandrosion	Meaning
Thales homothety	local inertial normalization	choose a scale where finite probes see a nearly flat metric
Riemann inversion	switch to a nonsingular spacetime chart	avoid a coordinate defect without changing the event
finite slope	causal segment difference	replace derivatives of the worldline by finite causal probes
equal-value pole barrier	caustic, chart, or curvature barrier	keep probes away from regions where finite information collapses
geodesic probe score	proper-time graph score	choose paths by an invariant action rather than by raw coordinates

Table 1: Dictionary from the earlier Pandrosion papers to the relativistic version. The physical invariant is not the chart; it is the proper-time functional and the causal structure.

This is the clean answer to the question “can Pandrosion replace Newton?”. Newton’s law is not replaced at the fundamental level; it appears as the weak-field, slow-motion limit of relativistic proper-time maximization. Pandrosion supplies the finite-probe mechanism that approximates that geometric law.

7 Newtonian limit

Restore c . In a weak static field with Newtonian potential Φ , use the standard first post-Newtonian metric model

$$ds^2 = - \left(1 + \frac{2\Phi}{c^2}\right) c^2 dt^2 + \left(1 - \frac{2\Phi}{c^2}\right) \|dx\|^2 + \mathcal{O}(c^{-4}).$$

For a slow path $x(t)$, $\|\dot{x}\| \ll c$, the proper time satisfies

$$\begin{aligned} d\tau &= \frac{1}{c} \sqrt{-ds^2} \\ &= \left[1 + \frac{\Phi(x)}{c^2} - \frac{\|\dot{x}\|^2}{2c^2} \right] dt + \mathcal{O}(c^{-4}) + \mathcal{O}(\|\dot{x}\|^4/c^4). \end{aligned}$$

Thus maximizing $\int d\tau$ is equivalent, up to an additive constant and a positive factor, to minimizing

$$\int \left(\frac{1}{2} \|\dot{x}\|^2 - \Phi(x) \right) dt.$$

The Euler–Lagrange equation is

$$\ddot{x} = -\nabla\Phi(x).$$

Theorem 7.1 (Newtonian shadow of Pandrosion relativity). *Let the hypotheses of Theorem 4.3 hold in a weak static field, and assume all selected graph paths remain in the slow-motion regime. Then the spatial projections of maximizing Pandrosion worldlines converge, after the weak-field expansion, to solutions of Newton’s equation*

$$\ddot{x} = -\nabla\Phi(x)$$

with the same endpoint data.

Proof. Theorem 4.3 gives convergence to a proper-time maximizing timelike geodesic. The weak-field expansion above converts proper-time maximization into the classical action principle with Lagrangian $\frac{1}{2} \|\dot{x}\|^2 - \Phi(x)$. Its Euler–Lagrange equation is Newton’s law. $\square$

8 Finite holonomy and curvature

Geodesic motion uses the metric as input. General relativity also determines the metric through curvature. A Pandrosion route to the field equation should therefore begin with finite curvature measurements, not with symbolic derivatives.

Let $\Pi_{h,\square}$ be finite parallel transport around a small loop $\square_h(u, v)$ spanned by two probe directions u, v . Define

$$R_h(u, v)w = \frac{\Pi_{h,\square}w - w}{\text{Area}_h(u, v)}.$$

If the transport rule is consistent with the Levi-Civita connection, then this is the natural Pandrosion estimator for the Riemann curvature tensor.

Conjecture 8.1 (Finite holonomy convergence). *On compact regions where the metric is C^3 and the finite transport rule is second-order consistent with Levi-Civita transport,*

$$R_h(u, v)w \longrightarrow R(u, v)w$$

uniformly for bounded probe directions as $h \rightarrow 0$.

Assuming this convergence, one can form finite Ricci and scalar curvature estimators by contraction:

$$\text{Ric}_h(u, v) = \text{tr}[w \mapsto R_h(w, u)v], \quad S_h = \text{tr}_g \text{Ric}_h,$$

and then the finite Einstein tensor

$$G_h = \text{Ric}_h - \frac{1}{2}S_h g_h.$$

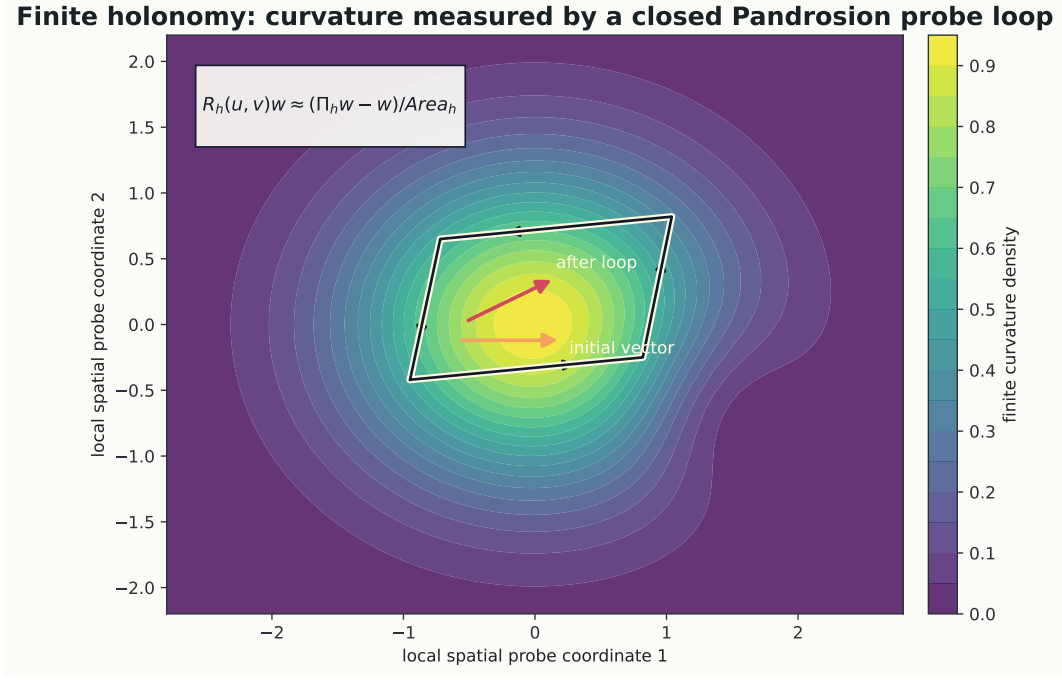


Figure 3: Finite holonomy curvature. Parallel transport around a small closed probe loop returns a vector with a finite rotation. Dividing the defect by the loop area gives a finite-slope curvature estimator.

Conjecture 8.2 (Pandrosion field balance). *A finite-probe formulation of the Einstein equation should take the form*

$$G_h + \Lambda g_h = \frac{8\pi G}{c^4} T_h + o(1),$$

where T_h is a finite-volume stress-energy estimator. If the holonomy, contraction, and stress-energy estimators are consistent, the limit is the classical Einstein equation

$$G + \Lambda g = \frac{8\pi G}{c^4} T.$$

This conjecture is deliberately conservative. It does not assert a new physical equation. It states that the Pandrosion way to make the equation finite is through holonomy loops, atlas conditioning, and causal probes.

9 Practical algorithm

A numerical Pandrosion-relativity engine can be written as a finite pipeline.

Step 1. Build a local event cloud X_h around the current event p_k .

Step 2. Remove endpoints outside the future cone, producing $\mathcal{C}_h^+(p_k)$.

Step 3. For each candidate, compute finite proper time, chart condition, tidal stiffness, and finite-slope acceleration defect.

Step 4. Select the next event or a short graph path by maximizing $\mathcal{J}_h$.

Step 5. Update the active chart by minimizing the atlas condition cost.

Step 6. Repeat until the target hypersurface or endpoint neighborhood is reached.

In flat Minkowski space, this engine selects straight timelike lines between fixed endpoints. In a curved spacetime, it selects discrete approximations to timelike geodesics. Near a coordinate singularity, it should switch charts rather than misreading the coordinate blow-up as a physical force.

10 What is new here

The formulas of relativity themselves are classical. The new Pandrosion contribution is the organization:

- (i) a causal finite-slope probe set replaces infinitesimal tangent vectors;
- (ii) a maximizing graph score replaces the differential geodesic equation;
- (iii) dynamic atlas conditioning replaces dependence on a single coordinate system;
- (iv) Thales-style scale normalization becomes local inertial normalization;
- (v) finite holonomy provides the curvature estimator needed for a future field-equation discretization.

This is enough to make a genuine research program: prove sharper convergence rates, test chart-selection rules near horizons and caustics, and compare finite-holonomy field balances with standard Regge calculus or finite-volume relativity.

11 Conclusion

Pandrosion relativity is best read as a finite geometric calculus for the standard relativistic principle of motion. It respects the causal cone, maximizes proper time for massive free particles, uses dynamic atlases to avoid coordinate artifacts, and converges to timelike geodesics under graph consistency. In the weak-field slow-motion limit it gives Newton’s law. The next frontier is not to claim a replacement for Einstein’s equation, but to build a finite-holonomy Pandrosion version of curvature and stress-energy that converges to it.

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